Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

3rd Year 2nd Semester Final Examination-2022

Session: 2019-20

Course Code: EEE 329 **Course Title:** Basic Communication Engineering

Total Marks: 100 **Credit: 3.0** **Total Time: 3 Hours**

Part-A

[Answer any two of the following questions]

1. (a) Write short notes on the following communication channels:

   (i) Twisted pair cable
   (ii) Radio wave 6

   (b) For carrier frequency 100 kHz, find the frequency, sketch the spectrum, indicate the upper sideband, lower sideband and their amplitudes of the following signals.

   (i) $m(t) = 1 + \cos(377t)$
   (ii) $m(t) = 2\sin^2(100\pi t)$
   (iii) $m(t) = 3\cos(1000t)\cos(3000t)$ 6

   (c) States Shannon's capacity theorem. A user wishes to transmit data over a channel having bandwidth B = 3000 Hz. If the SNR is 9.08 find the theoretical maximum bit rate of the channel. Also find the required SNR if the desired bitrate of the channel is doubled. 6

   (d) Find the equivalent noise resistance for several amplifiers in cascade. 7

2. (a) Show that the average information per message is equal to the entropy of the information source. 8

   (b) An angle modulated signal with carrier frequency 10 MHz is described by the equation $s(t) = 12\cos(\omega_c t + \sin 3000\pi t + 10\sin 2000\pi t)$

   (i) Find the power of the modulated signal
   (ii) Find the frequency deviation Δf
   (iii) Find the β
   (iv) Find the bandwidth. 8

   (c) Explain the effect of phase and frequency error during coherent detection of DSB-SC. 6

   (d) Describe the condition on message signal spectrum for SSB generation using selective filtering. 3

3. (a) Using proper block diagram explain phase shifting method of SSB generation. 8

   (b) Explain the effect of variation of amplitude and frequency of message signal on the spectrum and bandwidth of FM. 8

   (c) Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3 MHz and $\Delta f = 10.24\ kHz$. A NBFM generator of $f_{c1} = 20\ kHz$ and $\Delta f = 5\ Hz$ is available. Only frequency doublers can be used as multipliers. Additionally, a local oscillator (LO) with adjustable frequency between 400 and 500 kHz is readily available for frequency mixing. 9

Free shared copy · Nur Mohammad

Part-B

[Answer any two of the following questions]

4. (a) In an FM system the frequency deviation is 5 kHz, when its audio modulating frequency is 400 Hz, and the audio modulating voltage is 2 V. Compute the modulation index. Also compute the frequency deviation and the modulation index if (i) if voltage is increased to 6V keeping modulation frequency unaltered, (ii) if voltage is increased to 8 V while the modulating frequency is reduced to 200 Hz. 5

   (b) Explain slope detection method of FM demodulation. 6

   (c) Define and explain uniform and non-uniform quantization. 4

   (d) Five telemetry signals, each of bandwidth 2400 Hz, are to be transmitted simultaneously by binary PCM. The signals must be sampled at least 20% above the Nyquist rate. Framing and synchronizing requires an additional 0.5% extra bits. A PCM encoder is used to convert these signals before they are time-multiplexed into a single data stream. Determine the minimum possible data rate (bit per second) that must be transmitted, and the minimum bandwidth required to transmit the multiplexed signal. 5

   (e) Show that the receiver of DM is essentially an accumulator. 5

5. (a) Explain principle of sigma-delta modulator. Also explain how it is advantageous over DM. 5

   (b) Briefly explain the desired characteristics of a line coding scheme. Draw the line coded signal for the information bits '1010011100' for the following line coding scheme- i) Manchester, ii) AMI, iii) Di-code. 7

   (c) Explain DPSK principle with necessary block diagram. 8

   (d) Describe coherent detection of BFSK. 5

6. (a) With proper block diagram, describe modulation of QPSK. Also explain how it offers advantages over BPSK? 8

   (b) A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output for the following inputs- '110010', '001010', '101101', '000111'. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration? 6

   (c) Briefly explain frequency division multiplexing techniques. 5

   (d) Write short notes on following terms-

      (i) Pulse stuffing
      (ii) Multi-slot multiplexing 6

Free shared copy · Nur Mohammad